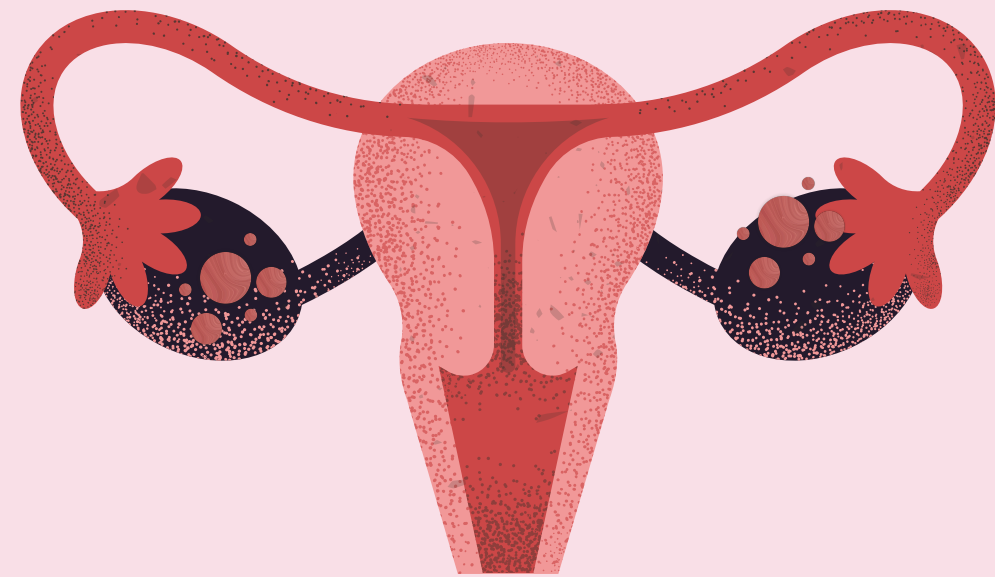


DATA-DRIVEN IDENTIFICATION OF PREDICTIVE FEATURE SETS FOR **PolyCystic Ovary Syndrome** CLASSIFICATION

Aleksandra Zečević
Mentor: Marko Tkalčič, PhD
UP FAMNIT, june 2026

INTRODUCTION & MOTIVATION



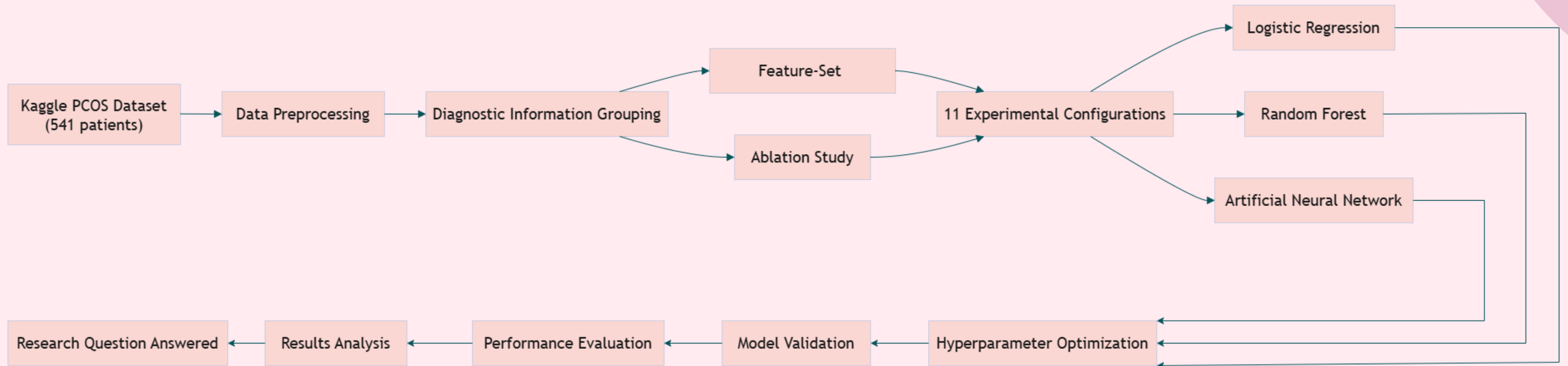
What is PCOS?

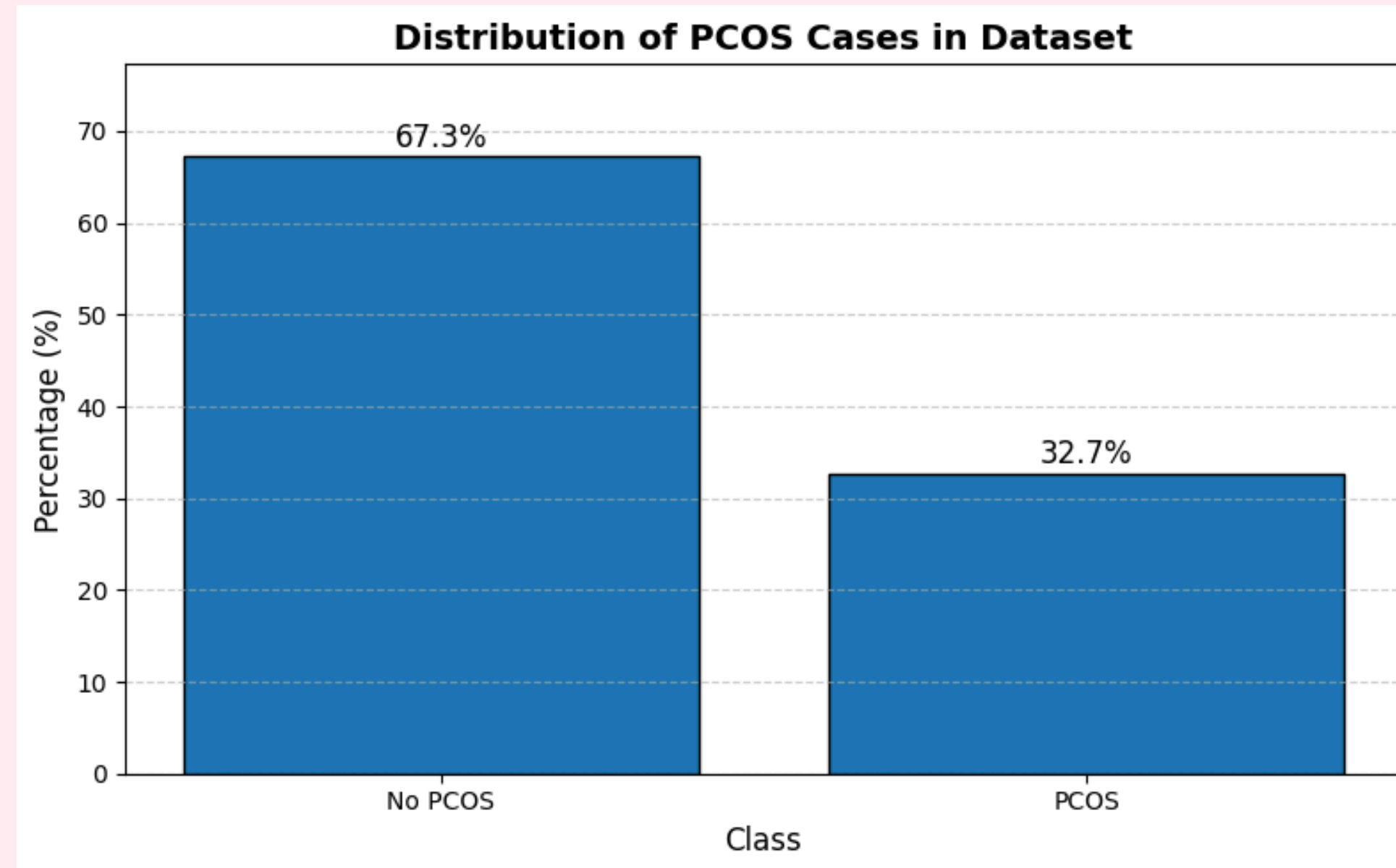
- Endocrine disorder
- Affects 10–13% of women
- Up to 70% undiagnosed
- Long-term health consequences

RESEARCH QUESTION

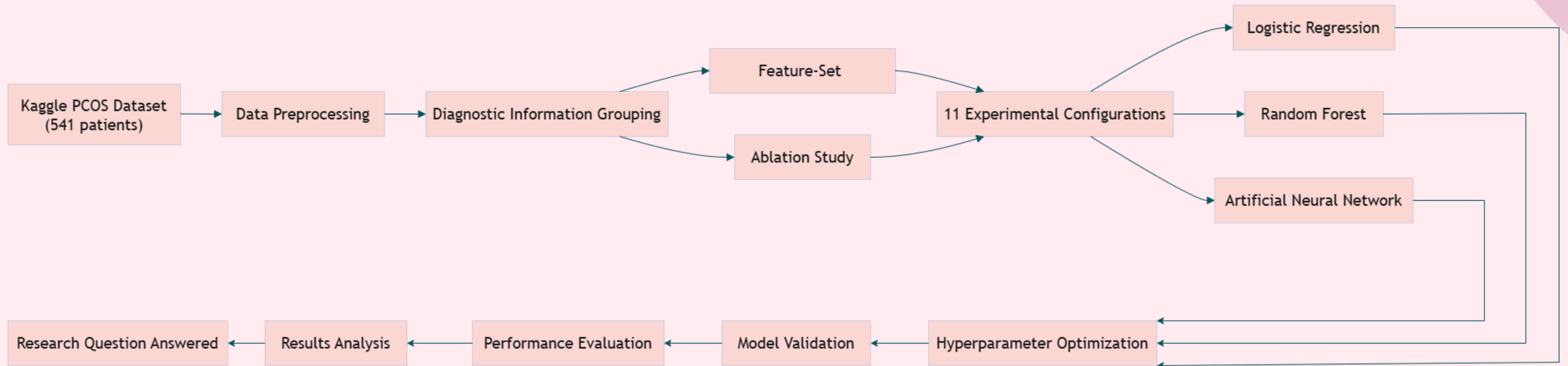
Which diagnostic information feature sets, including hormonal, blood-test, patient-observable, doctor-observable, and gynecological feature sets, provide the strongest predictive performance for PCOS classification?

METHODOLOGY





METHODOLOGY



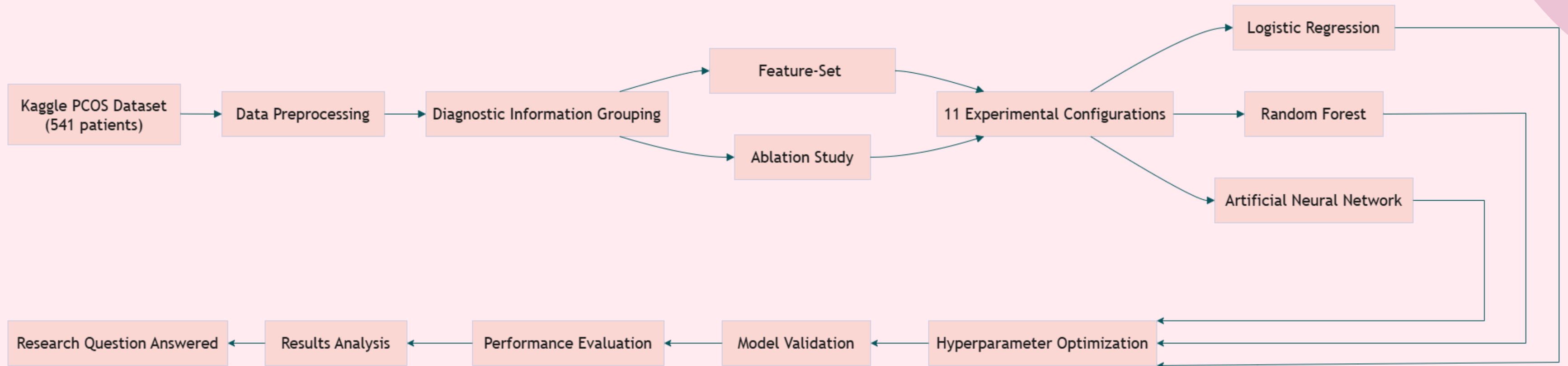
Feature Set Construction

All Features				
		Gynecological = Doctor + Ultrasound Measurements		
Blood test = Hormonal + Additional Lab Biomarkers			Doctor = Patient + Clinical Measurements	
Hormonal				Patient

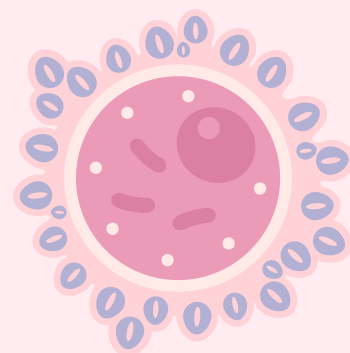
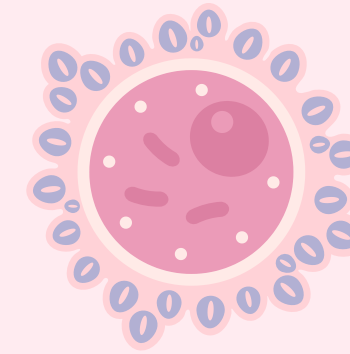
Ablation Study

- All minus Gynecological
- All minus Doctor
- All minus Patient
- All minus Blood test
- All minus Hormonal

METHODOLOGY



RESULTS



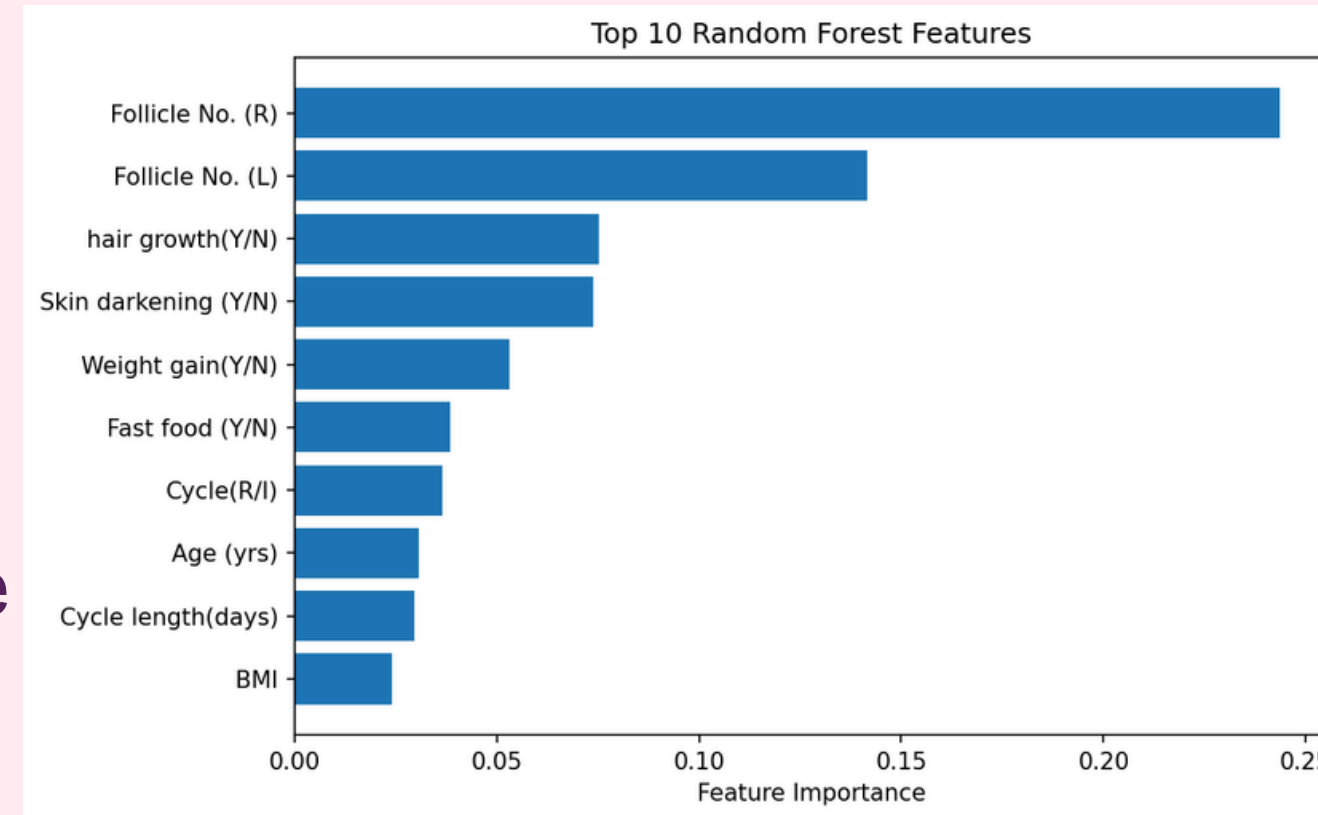
Strongest performing model for each incremental feature set

Feature set	Model	Acc	Rec	Prec	F1	CV Rec
Hormonal	RF	0.689	0.622	0.528	0.571	0.551
Blood-test	RF	0.667	0.622	0.500	0.555	0.533
Patient-observable	LR	0.815	0.804	0.698	0.748	0.771
Doctor-observable	ANN	0.807	0.761	0.700	0.729	—
Gynecological	RF	0.919	0.957	0.830	0.889	0.721
All features	RF	0.889	0.854	0.796	0.824	0.789

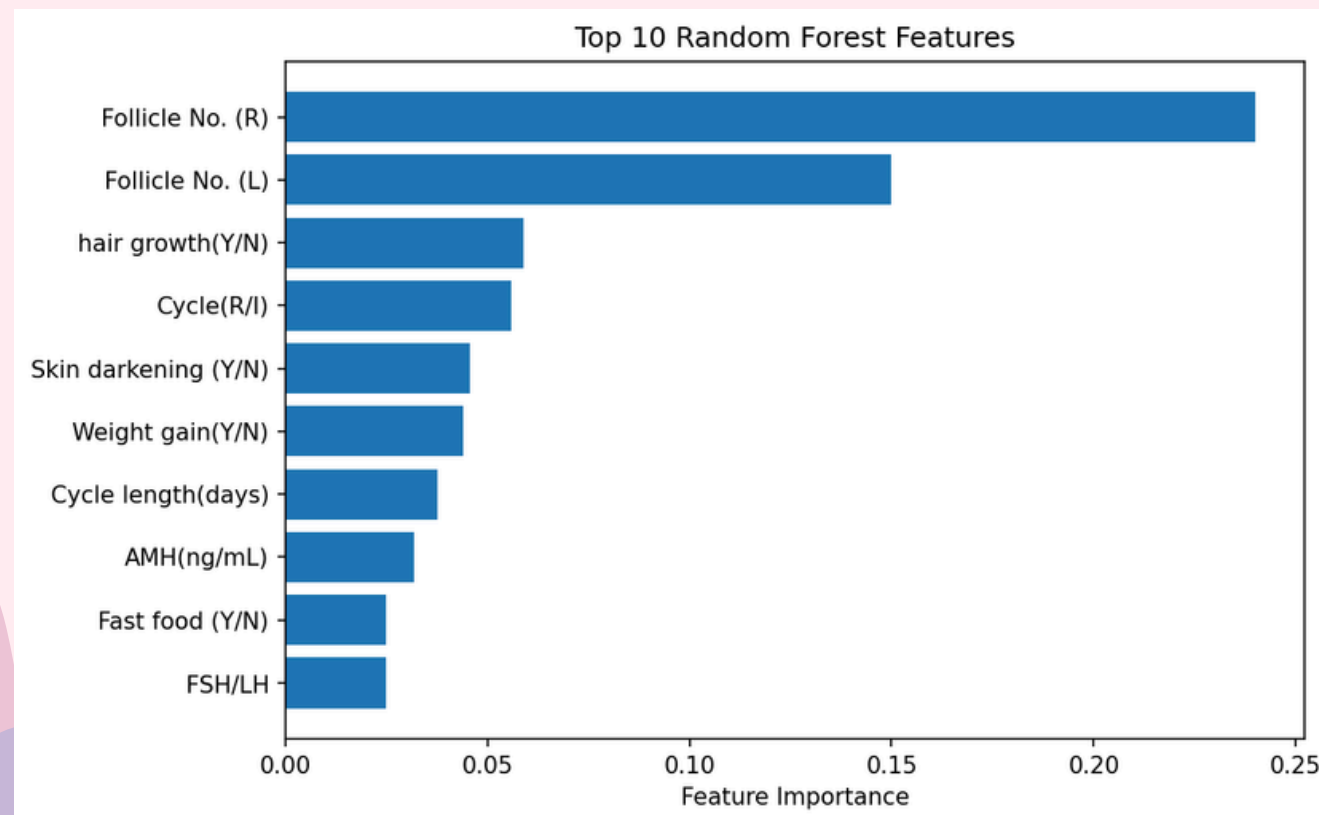
Strongest performing model for each ablation feature set

Removed feature set	Model	Acc	Rec	Prec	F1	CV Rec
Hormonal	RF	0.926	0.957	0.846	0.898	0.741
Blood-test	RF	0.919	0.957	0.830	0.889	0.761
Patient-observable	RF	0.844	0.778	0.761	0.769	0.616
Doctor-observable	RF	0.807	0.733	0.702	0.717	0.643
Gynecological	RF	0.659	0.578	0.491	0.531	0.542

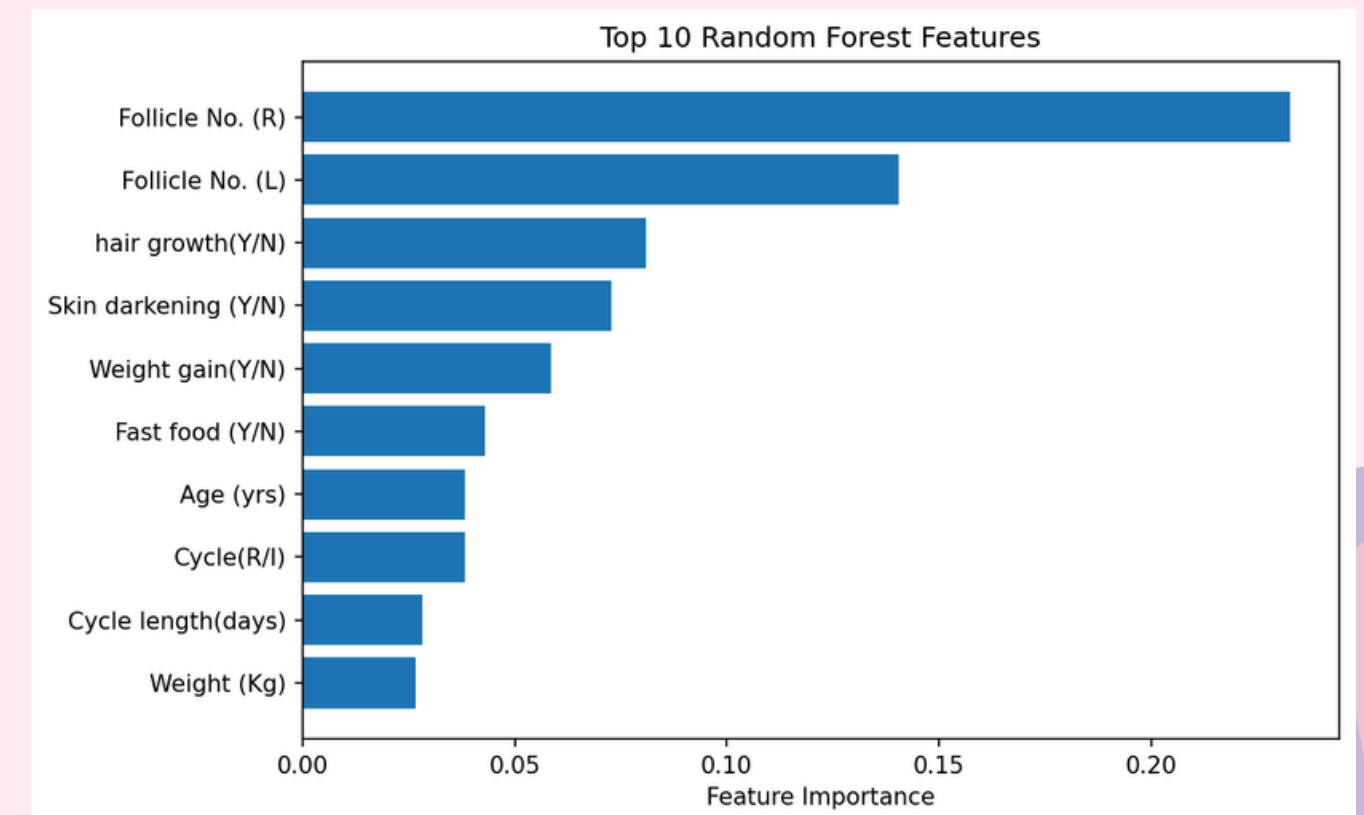
Feature importance scores for the all minus hormones set



Feature importance scores for the all features set



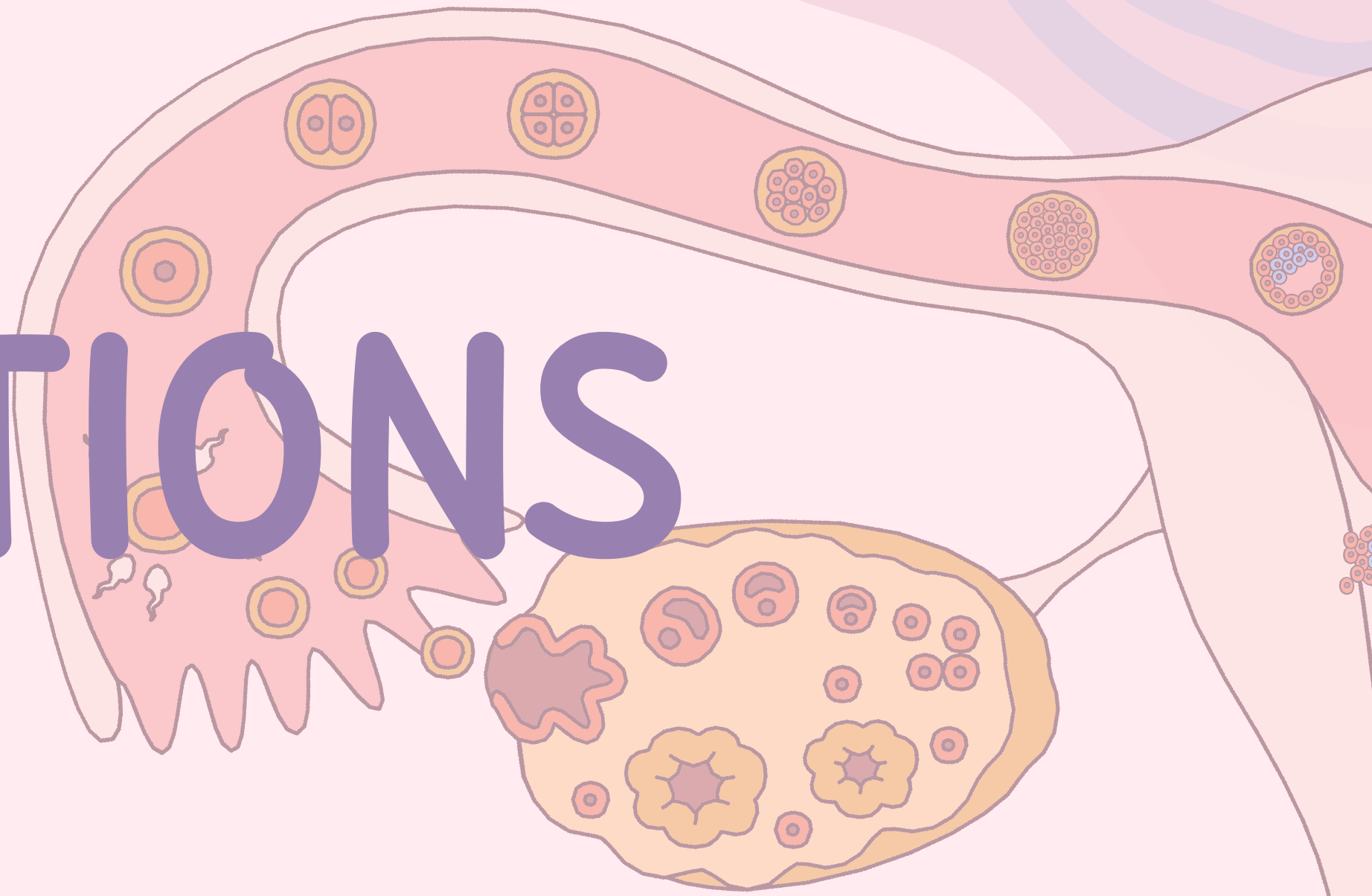
Feature importance scores for the gynecological set



CONCLUSION

- Gynecological features achieved the highest predictive performance.
- Diagnostic-information groups contribute unequally to PCOS classification.
 - More features do not necessarily improve performance.
 - Random Forest provided the most robust overall results.
 - The research question was successfully answered.

QUESTIONS?





THANK
YOU